Structural Formation of the Universe: As Observed by X-Ray Satellites

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Abstract— **Most celestial bodies emit their energy in the 0.5-5 keV range, thus the universe is very bright in X-rays. The Earth's atmosphere protects life on earth from being burned out by the high energy UV and X-ray, while blocking the precious information from the universe. Therefore, the X-ray satellites should be placed either at high altitudes in order to have a clear vision of stars, galaxies and unveil the cosmic mystery. According to structural formation hierarchy of the universe, stars form galaxies and galaxies form clusters. Galaxy clusters are the largest gravitational entities of the universe, and yet one of the brightest X-ray sources in the universe. In this paper, we review the large-scale structural formation of the universe based on the analysis results of our TÜBİTAK scientific research project on clusters of galaxies with X-ray data, which is obtained from the scientific satellites facing outer space such as NASA's Chandra, ESA's XMM-Newton and ISAS/JAXA's Suzaku.**

***Keywords*— Satellites; observation, galaxy, clusters, cosmology,**

I. Introduction

Clusters of galaxies are the largest stable entities and one of the brightest X-ray sources of the Universe. They are formed by gravitational collapse of galaxies and smaller groups. As they feel other and approach to form larger structures, galaxies eject significant amount of interstellar matter into the intergalactic medium, which is known as intra cluster medium (ICM). It is the ICM that glows and makes galaxy cluster observable in X-rays. The galaxies on the other hand dominate in the optical wavelengths. Most galaxies found in gravitationally bound groups. In 1937, Fritz Zwicky realized that the observable matter is not enough to explain the stability and balance of galaxies and clusters. About 80-85% of the matter is invisible, the so-called "dark matter".

In this work, we present the analysis results of several clusters of galaxies; from proto-clusters, regular hydrostatically stable clusters to super clusters of galaxies, which are composed of tens of clusters as a single unit. There are proto-clusters with very faint ICM (like Abell 194). These types of clusters are good cosmic laboratories to study hundreds of galaxies that subject to the same physical conditions. Therefore, we have studied several clusters to understand the nature of galaxies within ICM. The analysis results show that galaxies (AGNs) are bright at cluster's outskirts, which are the entrance gates of clusters' gravitational potential from the non-cluster field space. Our findings support that galaxies gets brighter when they fall into clusters. Due to ram pressure and gas stripping feeble balance of the active galactic nucleus and the surrounding gas is disturbed. The infalling gas fuels the AGN and/or awakes the dormant black holes.

After long enough time cluster dynamically evolves to a hydrostatical equilibrium stage unless an exceptional disturbance like merger of a subgroup or subcluster perturbs the structure. We have analyzed several relaxed cluster (etc A3571), but our analysis results show even at small-scale perturbations of temperature and metal distribution. This is mainly due to mergers; as cluster grows it attracts more matter into its potential. The intergalactic plasma is rich in metals, as we clearly observed from spectral analysis. The intra cluster matter is thought to be ejected from supernova explosions residing in member galaxies. We analyzed several SNR (like G156.2+5.2) and try to address the origin of cosmic metal abundances. Our finding support that type IA SNR are major chemical element suppliers of the clusters.

Recent observations and numerical simulations show that galaxies are not distributed in the Universe randomly. They are assembled in long filaments, web-like structures. The matter is spread into the space in the form of cosmic filaments and voids. In this work we try to map web like structures of the universe. We have selected 9 clusters of galaxies from Shapley supercluster region. The data is obtained from XMM-Newton (ESA), *Chandra* (NASA) and *Suzaku* (ISAS/JAXA). Flux and Luminosity relations are studied to explore possible interactions of intergalactic plasma and individual galaxies. X-ray brightness profiles of super cluster members show clear plasma elongation (e.g. A3560 in Shapley).

In this work, past, present and future of X-ray astronomy technology, satellites and missions will be covered, with a great emphasis on analysis results of large-scale structures of the Universe; galaxy clusters.

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Fig. 1 *Chandra* X-ray observatory (Courtesy: *Chandra*)

II. X-Ray Satellites

X-rays are emitted by hot plasma when charged particles are accelerated by high temperature or magnetic field. This emission mechanism takes place in interstellar halo, active galactic nuclei or intergalactic hot plasma. The atmosphere makes the earth habitable for life, while blocking the precious information of X-rays coming from celestial bodies. Therefore, the detectors should be placed near or beyond this blocking layer. After the first launch of X-ray satellite Uhuru in 1970, the X-ray astronomy has advanced rapidly within four decades. Similar to optical telescopes, parabolic surfaces are used in X-ray telescopes. Normal mirrors absorb X-ray photons. Thus, mirror surface should be coated with iridium, gold or platinum in order to reflect X-rays and focus on a CCD. The use of CCDs was introduced by Japanese satellite ASCA and followed by the current missions; *Chandra* (NASA), XMM-*Newton* (ESA) and *Suzaku* (JAXA). The Fig. 1 shows the technical sketch of Chandra X-ray observatory.

As in all fields of finance, future X-ray missions unfortunately under economical risk due to global crisis. NASA's Constellation-X mission and ESA/JAXA's XEUS mission are combined into a joint venture between NASA, ESA and JAXA as The IXO (international X-ray Observatory) project, which is planed for 2021 launch.

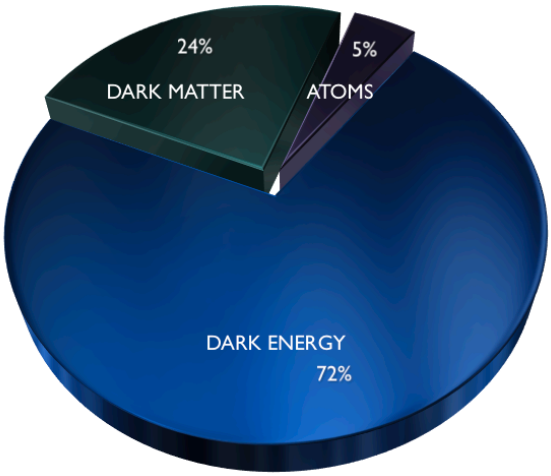

Fig. 2 Estimated distribution of matter and energy in the Universe

III. Structural Formation

The structural formation of the Universe started from a hot, dense, and nearly uniform state. However, sky is full of stars, galaxies, and larger structures of density clumps and voids in between which are not expected to be formed from a uniform distribution. Based on our current understanding of formation, 'cosmic inflation' is responsible at the early epoch for homogeneity and isotropy. After a while, nucleo-synthesis produces primordial elements. The Universe becomes a belnd of radiation and matter. Once matter dominates, gravitational instability and density perturbations build inhomogeneities. Consequently, matter falls into these dense regions.

The Universe is expanding with a changing rate since the beginning. This cosmic acceleration is explained by "dark-energy". The only way to know dark-energy is the expansion, and yet 72% of the Universe is in this form. 24% is in the form of dark-matter and only 5% is all we can observe in all possible wavelengths. In fig 2 we illustrate the estimated matter and energy distribution in the Universe.

A. Formation of a Galaxies

After recombination, density fluctuations increase or disperse and form filaments or voids, respectively. A lump of a dark matter and gas from time of recombination collapses under its own gravity and form a protogalaxy. There is still and ongoing debate about their formation. With the role of dark-matter in the hierarchy of structural formation, bottom-up model is widely accepted. Density perturbations and gravitational fluctuations appear as anisotropies in the cosmic microwave background (CMBR). COBE [1] and WMAP [2] observed these anisotropy and estimated CMB temperature of about 2.7 keV. By studying the fluctuations we can learn about the cosmic parameters as well as structural formation of the Universe.

Galaxies are the basic unit of cosmology. They contain stars, gas, dust and dark matter. The first protogalaxies are expected to be formed about 14 billion years ago. Billions of stars and the interstellar gas in between gravitationally bound and form galaxies. Two most distinct galaxy types are ellipticals and spirals.

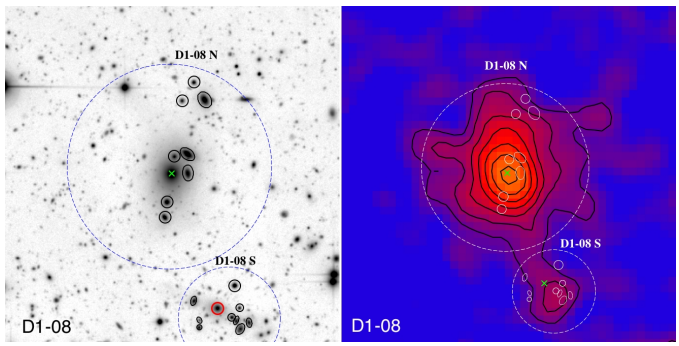

Fig. 3 Distant cluster hunting research from CFHTLS-D1field. Left: optical image, Right: X-ray image (courtesy: #108T226 project)

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B. *Formation of Groups & Clusters*

After their formation, galaxies still grow by interacting with other galaxies and form groups and clusters of galaxies. The clusters are the largest known gravitational entities of the universe. They are the intersection points of the cosmic filaments and thus the densest locations of the universe. Numerical simulations and observational evidences have shown that galaxies are not distributed evenly. Their positions form filaments and voids in the space separating them. These enormous structures somewhat took about 10 billion years to assemble, therefore it is rare to observe them in deep space.

In our current TÜBİTAK scientific research project #108T226 [3], we are searching for distant cluster candidates in CFHTLS deep fields. Fig.3 shows one of the galaxy cluster candidate in D1 field [3]. Left panel shows CFHTLS optical i-band image with encircled candidates of cluster members. Right panel is soft band X-ray emission from the same region obtained from XMM-Newton. The extend X-ray is emitted by the ICM of D1-08 cluster candidate.

C. *Formation of Filaments*

The matter is not homogenously distributed in the universe, but more like cosmic filaments and voids between them. Superclusters are the great walls of Universe. Galaxies are not spread randomly; they are located like pearls of a necklace. Fig. 4 shows the illustration of cosmic webs studied by Virgo consortium [4]. Right panel shows X-ray image overlaid the optical image was obtained by SuprimeCam at the Subaru telescope. The X-ray image showing the distribution of the diffuse hot gas (yellow to red) was obtained by XMM-Newton from the binary cluster of A222-A223 [5]. In our ongoing research project #109T092 [6], we are searching for the evidences of cosmic filaments and dark-matter within supercluster regions, in which close passages and mergers are more likely.

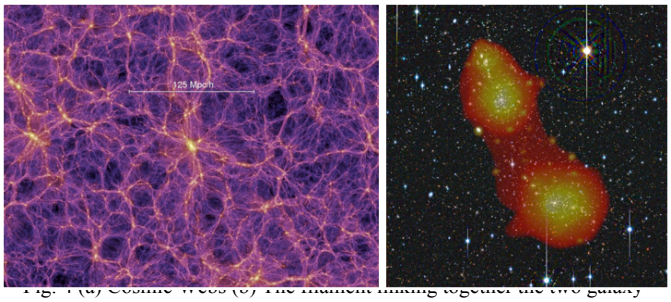

Fig. 4 (a) Cosmic webs (b) The filament linking together the two galaxy clusters, (credit: Virgo consortium, ESA/ XMM-Newton/ EPIC/ ESO)

IV. Clusters of Galaxies

The timeline of hierarchical formation of structures suggest that galaxies gravitationally gather to form clusters of galaxies. They are the one of the brightest extra galactic sources in the sky. They consist of ~100s of galaxies, with a typical size of 2-3 Mpc and a separation of ~10 Mpc. The intergalactic hot plasma has a temperature of 2-10 keV and metal abundance of 0.3-0.6 solar. They are composed of galaxies, intra-cluster medium and dark matter:

1) Galaxies: Galaxies preferentially found in groups or clusters. They dominate the optical picture of clusters. As they gravitationally gather together, they strip of their interstellar gas into the medium, which forms the extend hot ICM plasma.

2) Intra-Cluster Medium (ICM): The ICM is composed primarily of ordinary baryons. It is the very hot plasma mostly ejected and stripped matter from member galaxies. ICM strongly emits X-rays, which makes clusters visible. Group infalls and mergers heat ICM. Majority of the baryonic matter resides in the ICM, rather than galaxies and stars.

3) Dark Matter: Orbital velocities of galaxies in clusters are so high that it causes to a problem called 'missing mass'. Dark matter was first suggested by Firtz Zwicky in 1934 [7] in order to explain the velocities of galaxies in cluster potential.

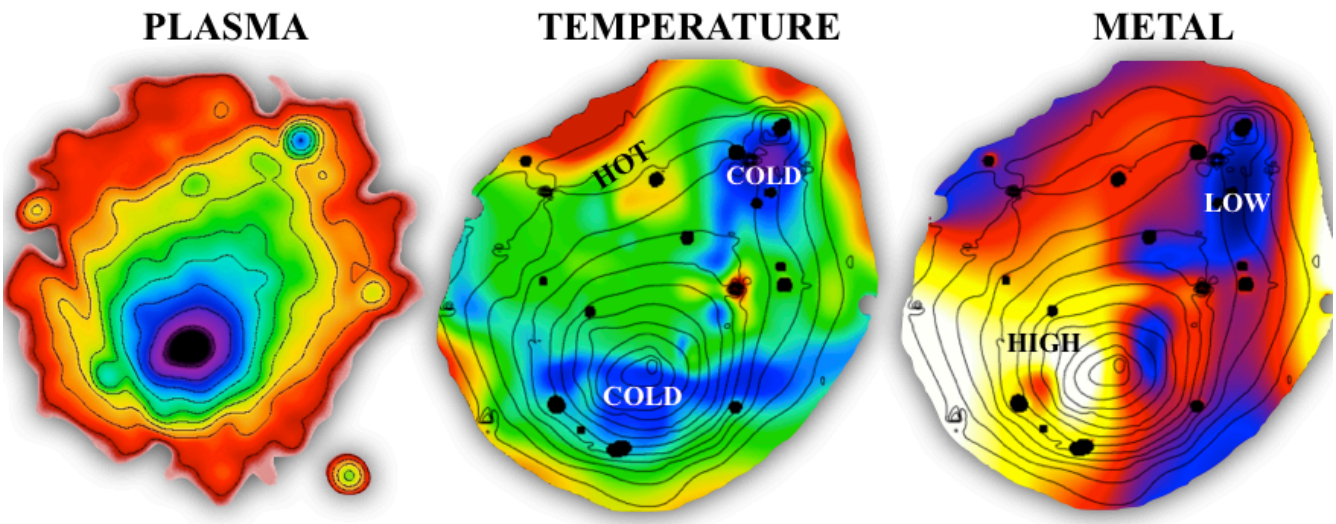

Fig. 5 Shapley member Abell 3560, (a) X-ray plasma, (b) temperature and (c) metal distribution maps [9] produced by X-ray data from XMM-Newton

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V. Cluster Dynamics

The observations and N-body simulations show that clusters of galaxies are dynamically violent environments. They evolve by infall of smaller galaxy groups and/or mergers of other clusters. The intergalactic hot X-ray plasma trapped in the gravitational potential well of clusters is strongly affected by dynamical processes. Mergers produce strong shocks, rapid rotation and turbulence, which can shock the intergalactic medium. Fig. 4 shows the recent analysis results from Abell 3560 from Shapley cluster. The temperature map is exhibited in the middle panel. The small-scale hot and cold plasma fluctuations are probably due to previous dynamical activities and unveil the morphological evidences of the cluster.

A. Mergers & Infalls

Clusters formed late in the structural formation timeline of the Universe. However, it is not the end of structural formation. The clusters further grow through mergers. The hot, X-ray bright, intergalactic plasma trapped in clusters gravitation is severely affected by mergers. Depending on the head-on or off-axis mergers, or the relative mass of colliding clusters, ICM is disturbed and shock heated. The ICM gets violent by strong shock, relativistic plasma, rapid rotation and turbulence motions. There are many on-going mergers, binary clusters or minor mergers of smaller scale disturbances in X-ray plasma (A3571 [8]). Fig. 4 displays elongation of X-ray plasma, and small-scale fluctuations of temperature and metal distribution of A3550 after mergers [9].

B. Chemical Enrichment of ICM

Intracluster medium (ICM) is dynamical environment continuously fed by matter ejected by SNRs and stripped gas from galaxies. ICM is also known as abundant in metalicity, which is ejected into the medium by supernova explosions. Hudaverdi spectrally confirms a clear metal ejection from cD galaxy of A1060 clusters of galaxies at his PhD thesis [10]. Notable ejecta are reported in the northern side of NGC 3311. North and south of NGC 3311 has a significant metal abundance difference. However, the spatial and spectral energy resolution is not educate to address the origin of the metals from the galaxy.

Recent studies show that the heavy elements are mostly due to supernovae type Ia [11]. We also in our *Suzaku* analysis of several SNR showed metal ejections [12]. Si and S are found to be mostly ejected elements, and thus it supports the previous findings that ICM is chemically enriched by type Ia supernovae.

Fig. 6 shows the X-ray isocontours from G348.5+0.1 [13] galactic SNR overlaid on radio image (green). The radio emission comes from relativistic particles, which are probably accelerated by supernova. The position of the radio rims supports the scenario of particle acceleration by the shocks. The red spots are the locations of Si and S abundant regions. The type of metals classifies the source as type 1a supernovae.

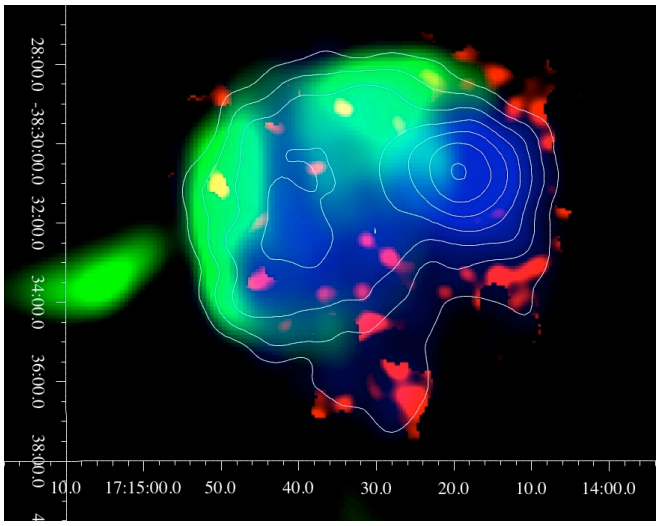

Fig. 6 G348.5+0.1 supernova remnant [13]. .

C. X-ray Plasma & Radio Interactions

Although the clusters are very significant in X-ray, violent merging activities bring the particles to the relativistic speeds. These particles lose their energy by emitting in radio wavelengths. Since relativistic particles cannot stay at high energy levels, the emission happens after a short while.

Radio emission from clusters of galaxies can be classified into halo and relic. Radio-halo is the large-scale radio emission mostly close to the centre. Radio relics are diffuse synchrotron emission coming from the outskirts and peripherals of clusters. Both large-scale areas of radio emission found in clusters of galaxies have no clear counterpart, unlike radio galaxy and AGN correlations. The driving mechanism and interactions of plasmas in different wavelengths is still a wonder.

In our research projects, we have studied several radio clusters, which showed promising implications of X-ray and radio plasma interactions. Fig. 7 displays our analysis results of a nearby faint cluster of galaxies, Abell 194. The cluster is also known as linear-cluster due to straight distribution of brightest members, as it is seen in Fig 6-a. A194 is very faint in X-rays and has an indefinite ICM, which allow us to observe individual X-ray galaxies clearly (Fig 6-b). The last panel in Fig. 6 shows radio emission from the same region. It is very interesting that, both northern and southern radio halos locate in very faint regions of X-rays, or in other words X-ray cavities. It is not clear whether strong radio emission pushes the X-ray plasma away or X-ray plasma cannot grow through radio plasma. The temperature map of A194 shows hot region ahead of radio relics. If this is not a chance location, it seems like radio jets sweeps X-ray plasma. The turbulent shocks heats the extend ICM in that part of the cluster.

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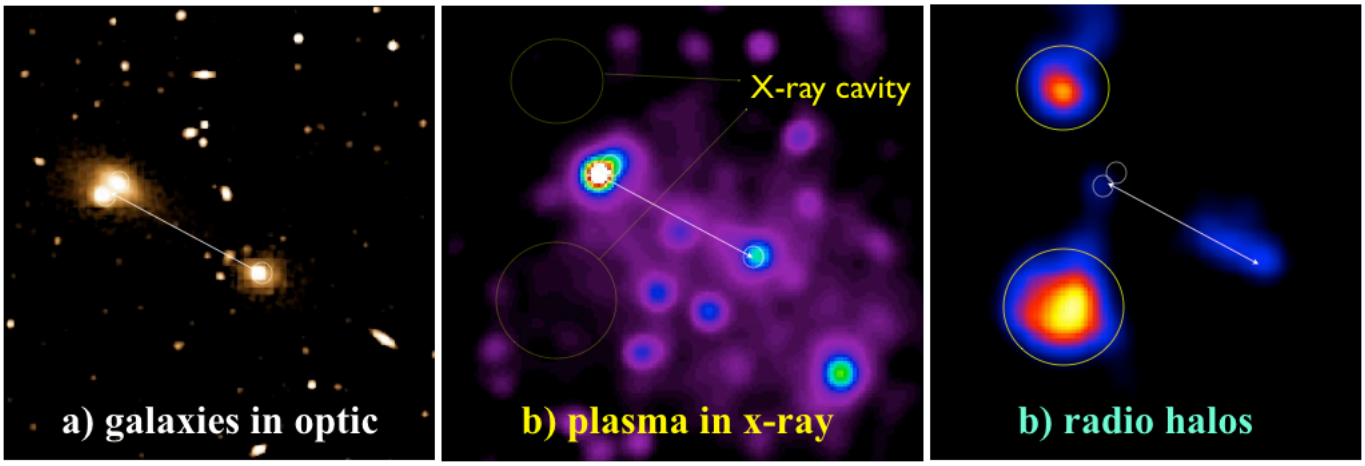

Fig. 7 Abell 194 linear clusters in (a) DSS optical image, (b) XMM-Newton soft band X-ray and (c) VLA radio image

D. Fate of AGNs

The galaxies are the most distinct members of the clusters. They are clearly visible in optical wavelengths and partly in X-rays. Because high density diffuse ICM burries most of the galaxies, detection becomes difficult. The galaxies from clusters and non-cluster fields indicate important results. Recent studies show that cluster galaxies are significantly brighter in X-rays [10]. When galaxies fall into the gravitational potential of a cluster and hits the high-density entrance gates at the outskirts, their physical balance is disturbed. The active galactic nuclei or dormant black hole awakes by fuelling with surrounding gas. As they further proceed into the centre they strip most of their fuel and cannot feed themselves. Thus, the bright galaxies are more likely to locate in the outskirts. The other explanation comes from the pairing of galaxies. Violent environment and high radial velocities in the core prevents galaxies to form bound-pair, which is believed to fuel AGNs.

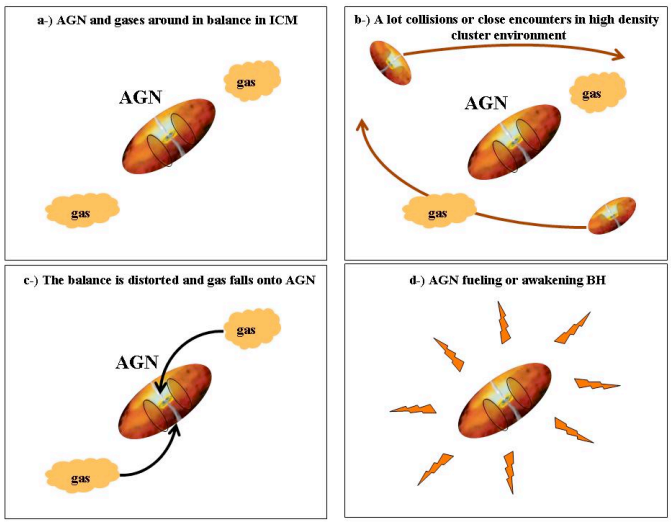

Fig. 8 The illustration of AGN fueling in the cluster outskirts.

VI. Superclusters

Galaxies gravitationally attract each other and form galaxy groups (~50) or galaxy clusters (>100). These large scale dynamics more likely to be observed at later phases of the hierarchical structural formation. Nevertheless galaxy clusters are not the ends of the formation. Clusters are further get together and form super clusters. Recent simulations and observations showed that universe forms cosmic filaments and cosmic voids based on the hierarchical formation of structure. The matter is not distributed randomly, but sponge like structure with sponge-walls and empty-balloons in between. Considering the edge of structural formation, no one has ever observed clusters of super-clusters. Their existence is still debating. Thus, super clusters are referred as End of Greatness.

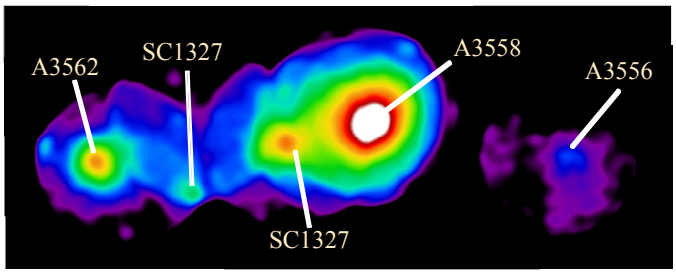

Fig. 9 X-ray image of Shapley core [credit #109T092 project].

Super clusters span to 100-million light-year width. If we consider a typical galaxy velocity of 1000 km/s, according to the Hubble law, a galaxy would travel only about 30-million light-year distance in 1/H (the age of the universe) time. This scale is quite small compared to super clusters size. This type of simple cosmic scale calculations always underestimates the observed expanding universe. Even the existence of super clusters gives the clues of structural formation of the universe. The matter is not homogeneously distributed, as expected. It forms cosmic filaments. Today's cosmic picture shows that universe is not dominated by regular matter, but dark-matter.

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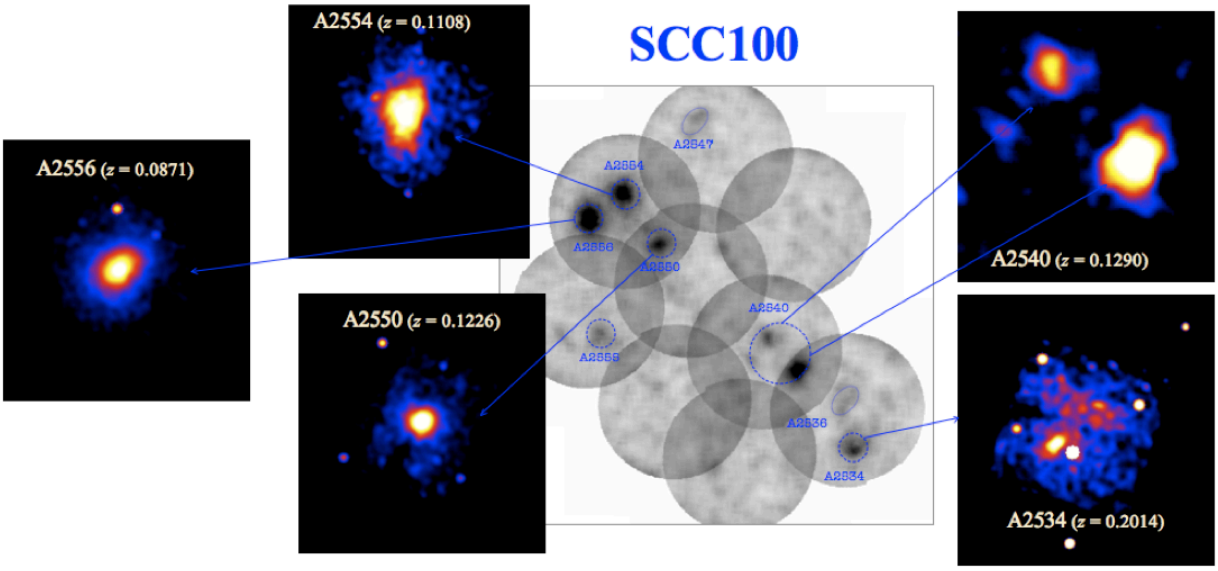

Fig. 10 Mosaic X-ray image of SCC100 region [credit #109T092 project].

In or ongoing project #109T092 entitled as "Mutual Interactions of Galaxy Clusters from Super Cluster Fields and Dark Matter Signatures from Cosmic Filaments" studies supercluster structures. In the project we analyze mainly 2 supercluster regions; 1- Shapley supercluster and 2- Aquarius (SCC100) supercluster regions. Fig. 9 and Fig. 10 show the mosaic composite X-ray images of two regions, respectively. Plasma elongations towards the close neighbors are notable. In this research we are looking for the observational evidences for dark-matter and its distribution from its gravitational effects on ordinary matter.

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